Exercises 4.2 1, 3, 7, 9, 13, 17, 19, 21, 25, 29, 33.

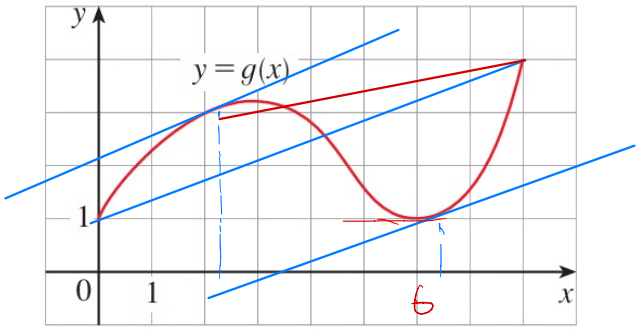
1. As can be seen from the graph, $f(x)$ is continuous on $[0,8]$

differentiable (?)

$f(0) = f(8)$ such that $f(x)$ satisfies Rolle's rule

$f'(1) = f'(5) = 0$

3. The graph of a function g is shown.



(a) Verify that g satisfies the hypotheses of the Mean Value Theorem on the interval $[0, 8]$.

(b) Estimate the value(s) of c that satisfy the conclusion of the Mean Value Theorem on the interval $[0, 8]$.

(c) Estimate the value(s) of c that satisfy the conclusion of the Mean Value Theorem on the interval $[2, 6]$.

a) ∵ $g(x)$ is continuous and differentiable at $[0,8]$.

By the MVT, there is a c, where $c \in (0,8)$.

$$f'(c) = \frac{f(8)-f(0)}{8-0} = \frac{4-1}{8} = \frac{1}{2}\ \frac{3}{8}$$

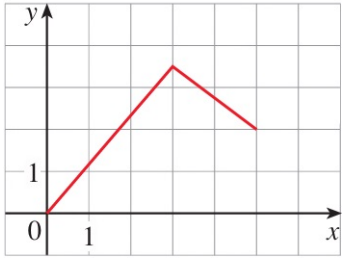
b) $x = 2.2$ or $x = 6.4$

c) $$f'(c) = \frac{f(6)-f(2)}{6-2} = \frac{1-3}{4} = -\frac{1}{2}$$

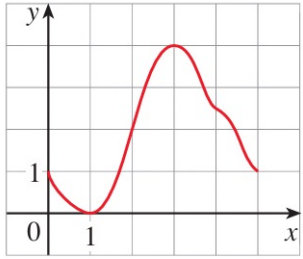
$c = 6$

5–8 The graph of a function f is shown. Does f satisfy the hypotheses of the Mean Value Theorem on the interval $[0, 5]$? If so, find a value c that satisfies the conclusion of the Mean Value Theorem on that interval.

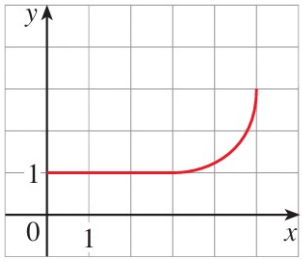
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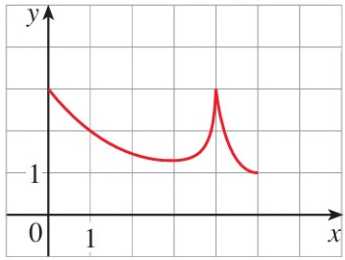
6.



7.



8.



13. Let $f(x) = 1 - x^{2/3}$. Show that $f(-1) = f(1)$ but there is no number c in $(-1, 1)$ such that $f'(c) = 0$. Why does this not contradict Rolle's Theorem?

$f'(x) = -\frac{2}{3}x^{-\frac{1}{3}} = -\frac{2}{3\sqrt[3]{x}}$

$f'(x)$ DNE at $x = 0$ such that $f(x)$ is not differentiable at $x = 0$

∴ Rolle's rule is not applicable.

So it doesn't contradict Rolle's rule since it's not applicable

15–18 Verify that the function satisfies the hypotheses of the Mean Value Theorem on the given interval. Then find all numbers c that satisfy the conclusion of the Mean Value Theorem.

15. $f(x) = 2x^2 - 3x + 1, \quad [0, 2]$

16. $f(x) = x^3 - 3x + 2, \quad [-2, 2]$

17. $f(x) = \ln x, \quad [1, 4]$

18. $f(x) = 1/x, \quad [1, 3]$

$f(x)$ is a logarithmic function so it's continuous on its domain $x \in (0, \infty)$

$f'(x) = \frac{1}{x}$ exists at $(1, 4)$. So $f(x)$ is differentiable at $(1, 4)$

$\therefore f(x)$ satisfies the Mean Value theorem

$$f'(c) = \frac{1}{c} = \frac{f(4) - f(1)}{4 - 1}$$

$$\frac{1}{c} = \frac{\ln 4 - \ln 1}{3}$$

$$\frac{1}{c} = 0.462$$

$$c = 2.165$$

19–20 Find the number c that satisfies the conclusion of the Mean Value Theorem on the given interval. Graph the function, the secant line through the endpoints, and the tangent line at $(c, f(c))$. Are the secant line and the tangent line parallel?

19. $f(x) = \sqrt{x}, \quad [0, 4]$

20. $f(x) = e^{-x}, \quad [0, 2]$

$$f'(x) = \frac{1}{2} x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$$

$$f'(c) = \frac{1}{2\sqrt{c}} = \frac{f(4) - f(0)}{4 - 0}$$

$$\frac{1}{2\sqrt{c}} = \frac{2 - 0}{4 - 0}$$

$$c = 1$$

21. Let $f(x) = (x - 3)^{-2}$. Show that there is no value of c in $(1, 4)$ such that $f(4) - f(1) = f'(c)(4 - 1)$. Why does this not contradict the Mean Value Theorem?

$f(x) = \frac{1}{x-3}$, domain $x \neq 3$ $\Rightarrow$

$f(x)$ is not continuous at $x = 3$

$f'(x) = \frac{-2}{(x-3)^3}$, $x \neq 3$

$f'(x)$ DNE when $x = 3$, such that $f(x)$ is not differentiable at $x = 3$

$\therefore$ the Mean Value theorem is not applicable for $f(x)$ in $(1, 4)$

$1^{-2} - (-2)^{-2} = \frac{-2}{(c-3)^3}$ $x \neq 3$

$1 - \frac{1}{4} = \frac{-6}{(c-3)^3}$

$\frac{3}{4} \times (-\frac{1}{6}) = \frac{1}{(c-3)^3}$

$c = 1$, which is not included in $(1, 4)$

This doesn't not contradict the mean value theorem since f is not continuous at $x = 3$, which is in the interval $x \in (1, 4)$

25. Show that the equation $x^3 - 15x + c = 0$ has at most one solution in the interval $[-2, 2]$.

let $f(x) = x^3 - 15x + c$ $\quad f'(x) = 3x^2 - 15$ and differentiable ✓

$\because$ $f(x)$ is a polynomial $\rightarrow$ $f(x)$ is continuous on $\mathbb{R}$

$f(-2) = (-2)^3 - 15 \times (-2) + c = 22 + c$

$f(2) = 2^3 - 15 \times 2 + c = -22 + c$

By IVT, in order to let at least one c, where $c \in (-2, 2)$, such that $f(c) = 0$,

$\begin{cases} f(-2) < 0 \\ f(2) > 0 \end{cases}$, which is not applicable

$\begin{cases} f(-2) > 0 \\ f(2) < 0 \end{cases} \Rightarrow -22 < c < 22$

Assume that there are 2 roots, where $a < b$ and both $\in (-2, 2)$

$f'(c) = \dfrac{f(b) - f(a)}{b - a} = 0$

$3x^2 - 15 = 0$

$3x^2 = 15$

$x^2 = 5$

$x = \pm\sqrt{5}$

$x = -\sqrt{5}$ is not belong to $x \in [-2, 2]$

Coordinate with MVT, the assumption of 2 roots are wrong

Therefore, $f(x)$ has one and only one root in $[-2, 2]$

29. If $f(1) = 10$ and $f'(x) \geqslant 2$ for $1 \leqslant x \leqslant 4$, how small can $f(4)$ possibly be?

$f'(c) = \frac{f(b) - f(a)}{b - a}$

$=$

$F(x) - G(x) = C$

33. Show that $\sin x < x$ if $0 < x < 2\pi$.

Mean Value theorem

let $0 < a < 2\pi$

$f(x) = \sin x$ is continuous and differentiable on $\mathbb{R}$

By MVT, there is a number c in $(0, a)$, such that

$$f'(c) = \frac{f(a) - f(0)}{a - 0}$$

$$\cos c = \frac{\sin a - 0}{a}$$

$$\sin a = a \cdot \cos c$$

$\because c < 2\pi \Rightarrow \cos c < 1$

$\therefore \sin a = a \cos c < a$

$\because a$ is an random value in $(0, 2\pi)$.

$\therefore \sin x < x$ for $x \in (0, 2\pi)$